Lifetime Asset Allocation in New Zealand: A Simulation and Optimization Approach

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1 Foreword and Research Context

This research is conducted in partnership with Consilium NZ LTD (referred to hereon as simply Consilium). Consilium is a New Zealand-based financial services company that provides a range of services to financial advisers and their clients. Their services include a range of tools and consulting services to support advisers in areas such as investment management, portfolio construction, risk management, and financial planning. Consilium also offers a range of investment products, including managed funds and portfolio solutions, that are designed to help advisers meet the needs of their clients. The goal of this research partnership is to examine and improve lifetime asset allocation strategies for individual investors, with a particular focus on retirees in the New Zealand context. This research aims to provide insights and recommendations that can help Consilium better serve their clients and support financial advisers in delivering exceptional financial advice.

While this research is conducted in partnership with Consilium, the views and opinions expressed in this paper are those of the author and do not necessarily reflect the official policy or position of Consilium. The author retains full academic independence in the conduct of this research, including the design, analysis, and reporting of findings. The author also notes that they have agreed with Consilium to retain full ownership of the intellectual property rights associated with this research, granting Consilium a non-exclusive license to use the research findings for their internal purposes and to share them with their clients and partners. The author notes that no confidential or proprietary information provided by Consilium has been used in the conduct of this research.

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2 Introduction

Managing a retiree's nest egg requires navigating two broad forms of financial risk. A well-established principle in investment research is that assets with higher expected returns tend to come with higher volatility (e.g. Campbell, 1996; Ghysels et al., 2005; Lundblad, 2007). Although many investors spend considerable energy selecting particular securities or funds, Ibbotson and Kaplan (2001) find that roughly 90% of the variation in fund returns is explained by the allocation to major asset classes such as equities, bonds, and cash. As a result, asset allocation is the primary driver of a portfolio's long-term behaviour. For the typical investor, whose retirement savings are usually held in managed funds that do not allow meaningful security selection, the most important strategic decision is choosing the overall exposure to these broad asset classes.

The central challenge for individuals approaching retirement is finding an allocation between safe and risky assets that provides the returns required for a comfortable lifestyle, while avoiding excessive exposure to loss. Achieving this balance is not straightforward. Once the allocation is considered as a function of risk appetite, savings capacity, spending needs, lifespan, and total accumulated wealth, the problem becomes complex. Many of these inputs differ significantly between individuals, and even small differences in behaviour or circumstance can lead to materially different outcomes. The mathematical problem of finding an optimal allocation through time is therefore highly sensitive to real-world considerations. This complexity has contributed to the emergence of various heuristics and rules of thumb that attempt to provide simple guidance in the absence of a fully individualised solution.

The significance of this problem extends beyond individual welfare. The global asset-backed pension market was valued at 68.9 trillion USD in 2024 (OECD, 2025). For many people, their retirement savings represent the largest financial asset they will ever manage apart from their home. Consequently, the question of how to allocate this wealth over the course of a lifetime is significant not only for individuals but also for governments. Even modest improvements in the efficiency of retirement strategies can reduce pressure on public pension systems. In New Zealand, for example, every resident aged 65 or older receives New Zealand Superannuation regardless of income or assets. The programme is funded through general taxation. Due to an ageing population and ongoing increases in life expectancy, the long-term fiscal sustainability of this system is becoming increasingly challenging. A recent Treasury report (The Treasury, 2025) estimates that spending on New Zealand Superannuation and healthcare could account for nearly half of all government expenditure by 2065 if current policy settings remain unchanged. This has led to suggestions that New Zealanders will need to save more for retirement, for example through higher KiwiSaver contributions, and that adjustments to eligibility age or means testing may be required. There is already a broad consensus that the public is under-saving for retirement (e.g. Dexter, 2025; Gilbert, 2025; Luxon, 2025; Palmer, 2025).

A critical requirement for the success of such policies is ensuring that individuals make appropriate decisions about their lifetime asset allocation. At present, the New Zealand government provides little guidance on how individuals should allocate their retirement savings. The Financial Markets Authority requires that default KiwiSaver funds adopt a Balanced risk profile (Financial Markets Authority, 2023), defined as holding between 35 percent and 63 percent growth assets (Financial Markets Authority, 2024). This replaced the previous Conservative default in 2021 (Retirement Commission Te Ara Ahunga Ora, n.d.), after it became clear that many KiwiSaver members did not switch funds from the default fund and that a conservative allocation was poorly aligned with the needs of long-term investors (Ministry of Business, Innovation and Employment and The Treasury, 2019). Although a Balanced fund is a reasonable one-size-fits-most solution, it is likely too conservative for many younger investors who have long investment horizons and the capacity to tolerate short-term volatility.

This paper has three objectives. First, we review the current literature on lifetime asset allocation, with particular attention to the glide-path strategies that dominate both academic discussions and industry practice. We also examine the different risk measures and optimisation frameworks that researchers have used to evaluate such strategies. Second, we analyse the spending needs and risk concerns of typical retirees, with a particular focus on the New Zealand context. Understanding these patterns is essential for designing realistic simulations of retirement behaviour and for evaluating the amount of risk that individuals can tolerate. Finally, we develop and test a flexible simulation and optimisation framework for lifetime asset allocation. Building on the work of Mäkinen and Toivanen (2024), who propose a Monte Carlo portfolio optimisation framework, we adapt and extend their method to address the specific challenges that arise in lifetime allocation problems. We also modify their risk measure so that it reflects the practical concerns of real-world investors.

Our aim is to show that this framework can generate stable and robust strategies, while remaining flexible enough to incorporate a variety of risk measures, asset return models, and spending assumptions. This should allow us to test the validity of existing glide-path recommendations and to provide a foundation for further research in this area of personal finance.

2.1 The Status Quo on Lifetime Asset Allocation

In order to understand the rationale for any allocation strategy, it is important to have a basic understanding of the financial life-cycle of the typical investor. Roughly speaking, an investor's life can be divided into two main phases: the accumulation phase and the decumulation phase. During the accumulation phase, which typically spans from early adulthood to retirement age, the investor focuses on building wealth through regular contributions to savings and investment accounts. During this period, the investor increases the wealth in their portfolio through a combination of contributions and investment returns. The decumulation phase begins at retirement and continues until the end of the investor's life. During this phase, the investor withdraws funds from their portfolio to cover living expenses and other financial needs. The goal during this period is to manage withdrawals and the investment's returns in a way that ensures the portfolio lasts throughout retirement. This two-phase framework also leads to a corresponding life-cycle wealth trajectory, in which wealth accumulates during the working years, peaks at retirement, and then might decline during retirement as funds are withdrawn. Although it is often assumed that this wealth trajectory is hump-shaped, with wealth declining steadily during retirement, to what extent wealth actually declines during retirement for many investors (or even whether it declines at all) is an open question that we will return to in a later section.

Another important concept in this discussion is human capital, which refers to the present value of an individual's future earnings potential. During the accumulation phase, human capital is typically high, as the investor has many working years ahead. As the investor approaches retirement, human capital declines, and at retirement, it effectively becomes zero. This shift in human capital has important implications for asset allocation decisions, as it affects the investor's overall risk profile and capacity to absorb investment losses.

As discussed earlier, genuinely optimal lifetime allocation depends on many individual factors. Despite this complexity, a large body of rules of thumb and financial products has emerged to guide investors. The most common is the declining-risk strategy, in which exposure to equities decreases with age. This idea has been influential for decades and was even considered as part of potential reforms to the United States Social Security system under the George W. Bush administration (R. Shiller, 2005; R. J. Shiller, 2006). A well-known example is the "age in bonds" rule, which suggests holding a percentage of the portfolio in bonds equal to the investor's age. This general approach appears throughout investment textbooks (e.g. Bengen, 1996; Bodie et al., 2023) and in the popular financial advice literature (Choi, 2022). Dolvin et al. (2010) review U.S. target-date funds and find that the majority follow a declining-risk structure that often resembles a 120 minus age equity allocation rule. In New Zealand, SuperLife's Age Steps product follows a similar pattern, reducing equity exposure from 95 percent at age step 20 to 10 percent at age step 80 (SuperLife, n.d.). Straightforwardly, the dominant paradigm in industry practice is to recommend that investors reduce their exposure to risky assets as they age through both the accumulation and decumulation phases. The intuitive logic behind declining-risk strategies is that older investors have less time to recover from losses, and that their spending needs become more immediate. In this view, reducing exposure to risky assets as retirement approaches is considered prudent.

However, the academic literature does not strongly support this approach, at least during the decumulation phase. Foundational work by Merton (1969) showed that when maximising consumption utility over a lifetime, the optimal asset allocation is constant through time and does not depend on age. Using empirical data, R. Shiller (2005) and R. J. Shiller (2006) found that declining-risk strategies may be inferior to a constant 100 percent equity allocation. Further studies reinforce this conclusion. Estrada (2014, 2015) find that declining-risk glide-paths tend to underperform both constant-risk and increasing-risk strategies across a wide range of risk measures, asset classes, and investor types. Using bootstrapped historical return data, they show that declining-risk strategies produce lower terminal wealth while providing no meaningful improvement in downside risk.

Declining risk strategies are somewhat less controversial during the accumulation phase (insofar as they support an equity-heavy allocation for younger investors), although even here the evidence is mixed. The most robust theoretical support arguably comes from Bodie et al. (1992) and Cocco et al. (2005), who show that when human capital is taken into account, younger investors should hold a higher proportion of equities. In their models, human capital is treated as a bond-like asset because it provides a steady stream of income that is less volatile than equity returns. As investors age and their human capital declines, their overall risk exposure increases, which justifies a reduction in equity holdings. In this model, they essentially argue that the declining-risk strategy is appropriate during the accumulation phase, insofar as it offsets the changing risk profile due to declining human capital.

However, work done on emprical data suggests that even during the accumulation phase, declining-risk strategies may not be optimal. Anarkulova et al. (2025) find that for typical young investors with low wealth and savings rates, a constant 100 percent equity allocation provides similar downside protection to a declining-risk strategy while delivering substantially higher expected returns. Using block-bootstrap resampled data from both U.S. and international markets, they find that optimised glide-paths still underperform portfolios invested entirely in equities, except in the very earliest years of retirement.

They argue that fixed-income assets offer very limited benefit once time-dependent return dynamics are incorporated, and that earlier studies tended to favour bonds because they assumed returns are independent and identically distributed from year to year. In their optimal glide-paths, bonds appear only modestly and mostly at the start of retirement, with allocations typically in the range of 30 to 40 percent or less. This produces a U shaped risk trajectory in which risky allocations dip at retirement and climb thereafter. This finding is consistent with Estrada (2015), who show that when the pre-retirement accumulation period is included, the optimal strategy often uses more bonds at the point of retirement before gradually increasing equity exposure again. The reasoning is that investors at the start of retirement have high accumulated wealth and almost no remaining human capital, which makes them especially vulnerable to sequence-of-returns risk, or the risk of experiencing a run of poor returns early in retirement. By holding some bonds at this point, the investor can reduce the likelihood of depleting their portfolio too quickly. However, once withdrawals reduce total wealth and the investor is further along in retirement, the impact of a run of bad returns becomes less severe, and therefore a higher equity allocation becomes more acceptable in fixed-spending scenarios.

In summary, while declining-risk strategies are widely recommended and implemented in practice, the academic literature does not strongly support their use, particularly during the decumulation phase. Instead, evidence suggests that constant or even increasing-risk strategies may provide better outcomes for retirees. While decreasing risk exposure may be appropriate during the final years of accumulation due to declining human capital, the benefits of such strategies appear limited when considering real-world investor behaviour and market dynamics. Where conclusions from the literature diverge, it often stems from differences in the underlying assumptions and methodologies used in the analyses. Particularly as it relates to the choice of risk measure, the modelling of asset returns, and the treatment of investor behaviour with regard to spending and withdrawals. For instance, the in the foundational work of Merton (1969), the find a constant asset allocation is optimal regardless of age, but this conclusion is based on the assumption of an investor who is withdrawing a fixed proportion of wealth each period. This result is similar to that of Anarkulova et al. (2025), who find that when the investor is modelled as making fixed-proportion withdrawals of 4% of wealth each year, a constant 100 percent equity allocation is optimal across the entire lifetime of the investor. However, this is simply not how most retirees behave in practice. Most retirees have a fixed consumption floor that they need to meet each year, regardless of the performance of their portfolio, and they have limited assets. Put simply, retirees cannot simply withdraw a fixed proportion of their wealth if their wealth is insufficient to meet their spending needs. This has important implications for the optimal asset allocation strategy, as it means that retirees are more vulnerable to sequence-of-returns risk and may need to adopt different strategies to manage this risk effectively. Additionally, many studies do not model an investor's changing spending needs over time. In practice, retirees may have different spending patterns throughout retirement, with higher life-style expenses in the early years and potentially lower regular expense, but likely higher healthcare costs, in later years.

The methods used to simulate asset returns also vary significantly between studies. Some researchers use historical return data, while others use bootstrapped or Monte Carlo simulated returns. The choice of return model can have a significant impact on the results, as different models may capture different aspects of market behaviour, such as volatility clustering or fat tails. For example, Anarkulova et al. (2025) argue that many earlier studies that supported declining-risk strategies relied on the assumption of independent and identically distributed returns, which does not accurately reflect real-world market dynamics. Real-world asset returns often exhibit serial correlation and changing volatility over time, which can affect the performance of different asset allocation strategies. In particular, sharp downturns in the market are often followed by fairly rapid recoveries (Dimensional Fund Advisors, 2023), which can benefit investors who maintain higher equity exposure during these periods.

Implications for Consilium NZ LTD

While Cosilium does not currently offer a glide-path product, nor do they explicitly recommend a declining-risk strategy, the findings from the existing literature have important implications for their business. It suggests that the current common-knowledge around lifetime asset allocation may be misguided, and that there is an opportunity for Consilium to differentiate themselves by offering more sophisticated and evidence-based asset allocation strategies. In particular, there may be an opportunity to challenge the assumptions of many of their advisors around need for declining-risk strategies.

Consilium has typically based much of its modelling of portfolio performance on IID return assumptions, which may not accurately reflect real-world market dynamics. By investigating asset allocation strategies that account for more realistic return dynamics, Consilium can provide more robust and effective advice to their clients.

2.2 Methods for Simulating Lifetime Wealth Trajectories

A key component of studying lifetime asset allocation is the ability to simulate realistic wealth trajectories for individual investors. This involves modelling both the accumulation and decumulation phases, taking into account factors such as contributions, withdrawals, investment returns, and spending needs. Our approach builds on a Monte Carlo simulation framework, which allows us to generate a wide range of possible outcomes based on different asset allocation strategies and market conditions. Importantly, the smulation approach we propose is agnostic to the source of asset returns, meaning that we can use historical data, bootstrapped returns, or simulated returns from a variety of models. This flexibility allows us to test the robustness of different asset allocation strategies across a range of assumptions. Additionally, the framework also allows for the incorporation of deposits and withdrawals of different amounts at different times, which is a key feature of real-world investor behaviour.

Unlike the approach proposed by Merton (1969), ours is a discrete-time simulation that models the investor's wealth at regular intervals. This allows us to capture savings and spending behaviour in fixed-dollar amounts, which is more realistic for most retirees.

Each step of the model takes the following form:

$$W_{t+1} = g_{t+1} W_t + \pi_{t+1} \tag{1}$$

Where W_t is the wealth at time t, g_{t+1} is the growth factor on the portfolio between time t and $t+1$, and π_{t+1} is the net contribution (or withdrawal if negative) made at time $t+1$. This can represent regular contributions during the accumulation phase or withdrawals to cover spending needs during the decumulation phase. The model can be easily adapted to include varying contribution and withdrawal amounts over time, allowing for a more nuanced representation of investor behaviour, including simulating variable or stochastic spending patterns.

The growth factor g_{t+1} is determined by the asset allocation strategy being tested, which specifies the proportion of wealth invested in different asset classes. Specifically, we define asset allocation as a vector $\mathbf{a}_t$ at time t, where each element $a_{i,t}$ represents the proportion of wealth invested in asset class i. The portfolio growth factor is then calculated as:

$$g_{t+1} = 1 + \mathbf{a}_t^\top \mathbf{r}_{t+1} \tag{2}$$

Where $\mathbf{r}_{t+1}$ is a vector of returns for each asset class between time t and $t+1$.

In a Monte Carlo simulation, we generate a large number of possible return paths. This means that for each time step, we have N different return vectors for our k asset classes. For each path, we simulate the investor's wealth trajectory over time using the equations above. This means that equation 1 has the matrix form:

$$\mathbf{w}_{t+1} = \mathbf{w}_t \odot \mathbf{g}_{t+1} + \boldsymbol{\pi}_{t+1} \tag{3}$$

Where $\mathbf{w}_t$, $\mathbf{g}_{t+1}$, and $\boldsymbol{\pi}_{t+1}$ are vectors of length N representing the wealth, portfolio returns, and contributions or withdrawals for each of the N simulated paths at time t, and $\odot$ denotes element-wise (Hadamard) multiplication.

The growth factors $\mathbf{g}_{t+1}$ are determined by the asset allocation strategy and a $N \times k$ matrix of returns $\mathbf{R}_{t+1}$ for the k asset classes across the N simulated paths. The asset allocation strategy can be defined in various ways, the first of which is as a function of time (or age). In this case, we define the asset allocation at time t as $\mathbf{a}_t = f(t; \boldsymbol{\theta})$, where f is a function parameterised by a set of parameters $\boldsymbol{\theta}$.

This yeilds the portfolio growth factors as:

$$\mathbf{g}_{t+1} = 1 + \mathbf{R}_{t+1} \cdot f(t+1; \boldsymbol{\theta}) \tag{4}$$

In this form, the model can accommodate a wide range of asset allocation strategies based on the investor's age. This is similar to the approach that Anarkulova et al. (2025) use, and our first objective will be to replicate their findings for the optimal glide-paths, and to examine how these findings change when we modify the risk measures and spending assumptions to better reflect real-world investor behaviour, particularly in the New Zealand context.

Implications for Consilium and an Aside on Using Matrix Operations and Computational Efficiency

The effort taken here to express the model in matrix form is not only for academic rigor, but it also allows us to leverage vectorised operations through highly optimised linear algebra libraries (e.g. `numpy` (Harris et al., 2020)) which can significantly speed up the simulation process, especially when dealing with a large number of paths and time steps. A mistake made in previous work by Consilium when it has considered using this type of Monte-Carlo approach to simulating investor outcomes is that they have constructed very poor computational implementations that rely heavily on for-loops and other non-vectorised operations, which can be extremely slow and inefficient. By contrast, using matrix operations allows us to perform calculations on entire arrays of data at once, leading to many orders of magnitude improvement in speed. This work alone, if applied to Consilium's existing offerings would represent a significant improvement in the service it provides to its clients.

The use of matrix operations will also make it easier when it comes to optimising the asset allocation strategy, as we can leverage automatic differentiation tools (e.g. `tensorflow` (Martín Abadi et al., 2015), `pytorch` (Paszke et al., 2019)) to compute gradients of the risk measure with respect to the parameters of the asset allocation strategy.

2.2.1 Wealth-Responsive Spending Policy

Rather than having a fixed spending pattern, we can also model the investor's spending as a function of their current wealth level. In some cases, retirees may have a fixed spending floor that they need to meet each year, regardless of the performance of their portfolio. In other cases, retirees may adjust their spending based on the performance of their portfolio, for example by reducing discretionary spending during periods of poor returns. To model this behaviour, we can define the net contribution (or withdrawal) at time t as a function of wealth: $\pi_t = g(W_t; \boldsymbol{\phi})$, where g is a function parameterised by a set of parameters $\boldsymbol{\phi}$. This allows us to simulate more realistic spending patterns that can vary over time and in response to changes in the investor's wealth. For example, we could model a spending policy that has a fixed floor and a variable discretionary component that adjusts based on the investor's current wealth level. This could be represented as:

$$\pi_t = -\max(S_{\text{floor}}, \alpha W_t) \tag{5}$$

Where S_{floor} is the fixed spending floor, and α is a parameter that determines how much of the investor's wealth is allocated to discretionary spending.

Additionally, we can prevent the investor from withdrawing more than their current wealth by defining the net contribution as:

$$\pi_t = \max(-g_{t+1} W_t, -\max(S_{\text{floor}}, \alpha W_t)) \tag{6}$$

Equally, we could model a spending policy that has a random component, for example by adding a stochastic term to the spending function modelling rare, but significant, unexpected expenses that retirees may face, such as healthcare costs or home repairs.

2.2.2 Wealth-Responsive Allocation Policy

Like with the spending policy, we can also model the asset allocation strategy as a function of both time and wealth. In the age-based glide-path framework, the hypothetical investor's asset allocation strategy is only responsive to time (i.e. age), and does not take into account the investor's current wealth level. In practice, an investor's risk tolerance and spending needs may vary significantly depending on their accumulated wealth at any given time. For example, an investor with a large portfolio may be able to tolerate more risk than an investor with a smaller portfolio, even if they are the same age. To address this limitation, we will attempt to extend the framework using a method similar to that proposed by Mäkinen and Toivanen (2024), who introduce a control matrix that allows the asset allocation strategy to depend on both time and wealth.

In this framework, the asset allocation at time t and wealth W_t is given by $\mathbf{a}_t = C(t, W_t; \boldsymbol{\theta})$, where C is the control matrix parameterised by $\boldsymbol{\theta}$. This yeilds the portfolio growth factors as:

$$\mathbf{g}_{t+1} = 1 + \mathbf{R}_{t+1} \cdot C(t+1, W_t; \boldsymbol{\theta}) \tag{7}$$

The way that this control-matrix works is that it defines a grid over the time and wealth dimensions, with each cell in the grid corresponding to a specific asset allocation. During the simulation, the investor's current age and wealth level are used to determine which cell in the grid they fall into, and thus the corresponding allocation to the risky asset (the remainder of the portfolio is allocated to the safe asset).

This approach allows for a more nuanced asset allocation strategy that can adapt to the investor's changing circumstances over time. For example, an investor who has accumulated significant wealth may be able to take on more risk, while an investor with lower wealth may need to adopt a more conservative approach. However, it also exponentially increases the number of parameters that need to be optimised, which can make any optimisation process more challenging.

However, we can reduce the search-space somewhat by using less granular grid points for the control matrix and interpolating between them to determine the asset allocation for time steps and wealth levels that fall between the defined grid points. This allows us to maintain a reasonable level of flexibility in the asset allocation strategy while keeping the number of parameters manageable. Moreover, it is unlikely that investors' asset allocation policy will be meaningfully different for small increments in time or wealth, so using a coarser grid with interpolation is unlikely to significantly reduce the optimality of the resulting strategy, while making the optimisation process more tractable. This idea of interpolating between grid points was first proposed by Mäkinen and Toivanen (2024), who use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined wealth-grid points, however, we also propose that interpolation could be used for time steps that fall between the defined time-grid points as well, which would further reduce the number of parameters that need to be optimised.

To understand how this works on a more practical level, consider figure 1, which shows a simple example of a control matrix for a two-asset portfolio with allocations defined at 25 percent increments across a 5 year period. Our control matrix has 6 time steps (0 to 5) and 5 wealth levels (0 to 4), so is a 6 by 5 matrix. Each cell in the matrix corresponds to a specific allocation to the risky asset, with the remainder allocated to the safe asset. For example, if an investor is at time step 2 and has a wealth level of 3, they would fall into the cell corresponding to an allocation of 50 percent to the risky asset and 50 percent to the safe asset. For timesteps or wealth levels that fall between the defined grid points, we can use interpolation to determine the appropriate asset allocation. For example, if an investor is at time step 2.5 and has a wealth level of 3.5, we could use bilinear interpolation to calculate the asset allocation based on the four surrounding grid points.

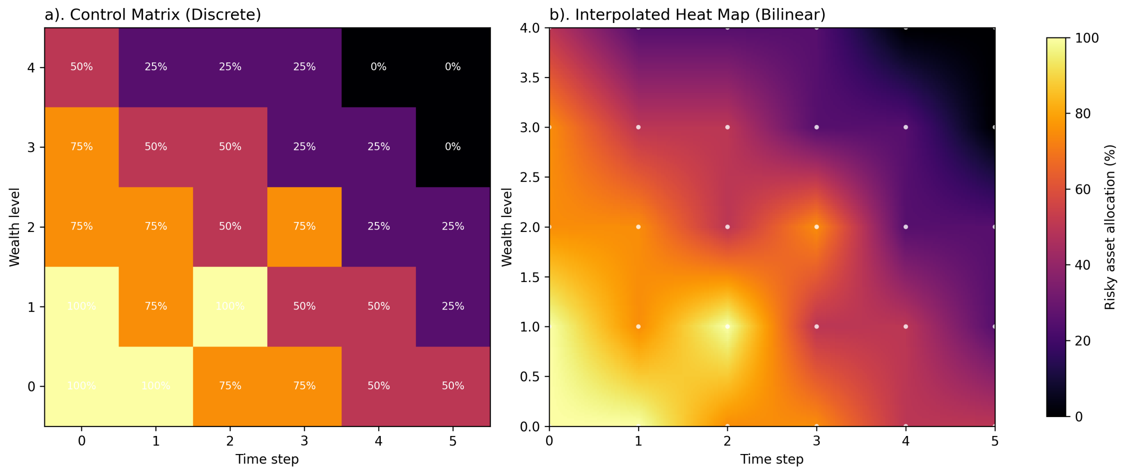

Figure 1: Illustrative example of a discrete control matrix (a) and bilinearly interpolated allocation surface (b).

For allocations with more than two asset classes, the control matrix can be extended to a higher-dimensional tensor, with each additional dimension representing an additional asset class. Mäkinen and Toivanen (2024) use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined grid points, ensuring a smooth transition between different allocation strategies.

2.3 Optimisation of Asset Allocation Strategies

2.4 Lifecyle Risk Measures

An essential component of optimising asset allocation strategies is the choice of risk measure. The risk measure serves as the objective function that we seek to minimise (or maximise) when searching for the optimal strategy. The choice of risk measure can significantly influence the resulting optimal strategy, as different risk measures capture different aspects of risk and may lead to different trade-offs between return and risk.

Our risk measure should be both interpretable and relevant to the concerns of real-world investors. It should be both easy to understand and directly related to the outcomes that investors care about. Additionally, depending on the optimistation method we choose, the risk measure may also need to be differentiable with respect to the parameters of the asset allocation strategy, which would allow us to use gradient-based optimisation techniques.

Traditional portfolio optimisation, like that proposed by Markowitz (1952), typically uses variance of asset returns as the risk measure. This idea is based on the assumption that investors are risk-averse and that they prefer portfolios with lower variance for a given level of expected return. However, when we consider lifetime asset allocation, this type of measure makes little sense as we are principally concerned with lifetime consumption and the risk of not being able to meet spending needs, rather than the variance of returns.

This idea does persist, a version of it makes an appearance in Mäkinen and Toivanen (2024), who demonstrates their method using variance and semi-variance of terminal wealth as the risk measure for optimising their control-matrix asset allocation strategy. This leads to some perplexing results, such as the optimal strategy being to risk-off above a certain wealth level, which is not something that would be expected to be optimal for a real-world investor, but does make sense when the optimiser is "targeting" a certain variance of terminal wealth.

Additionally, our risk measure should also be applied accross the entire lifecycle of the investor, rather than just at the terminal wealth.

2.4.1 What are the Concerns of Retirees?

It is clear that in order to construct a risk measure that is both interpretable and relevant to real-world investors, we need to have a good understanding of the concerns that typical New Zealand retirees have about their financial security in retirement. Unfortunately, there are is limited research on this topic in the New Zealand context, and so we will need to draw on research from other countries, particularly the United States, to inform our understanding of retirees' concerns.

The key challenge in optimising the asset allocation strategy is that the the parameter space can be very large, even for relatively simple strategies. For example, if we use are searching for an optimal age-based glide-path with 2 asset classes, with allocations in 10 percent increments across a 60 year period, this results in 11 possible allocations at each of 60 time steps, resulting in a total of 11^{60} possible strategies. This is clearly an intractably large search space, and so we must use numerical optimisation techniques to find a good solution.

Merton (1969) use a formulation where the investor maximises a utility function over their lifetime, based on the consumption at each time step. This approach has the advantage of being mathematically tractable, but it relies the investor consuming a fixed proportion of their wealth at each time step, which is not realistic for most retirees. Additionally, its closed-form solution assumed that asset returns are IID, which is the type of return dynamics that Anarkulova et al. (2025) argue leads to misleading conclusions about the optimality of declining-risk strategies, and so this approach is not suitable for our purposes.

Anarkulova et al. (2025) use a recursive approch where a grid-search is used to find the optimal allocation at each time step, working forwards in passes until convergence is reached. This approach will likely work for the age-based glide-path optimisation, but it is not clear how it would extend to the control-matrix approach, where the asset allocation depends on both time and wealth, as this increases the number of parameters by an order of magnitude again.

In their paper, Mäkinen and Toivanen (2024) optimised their control-matrix asset allocation strategy

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